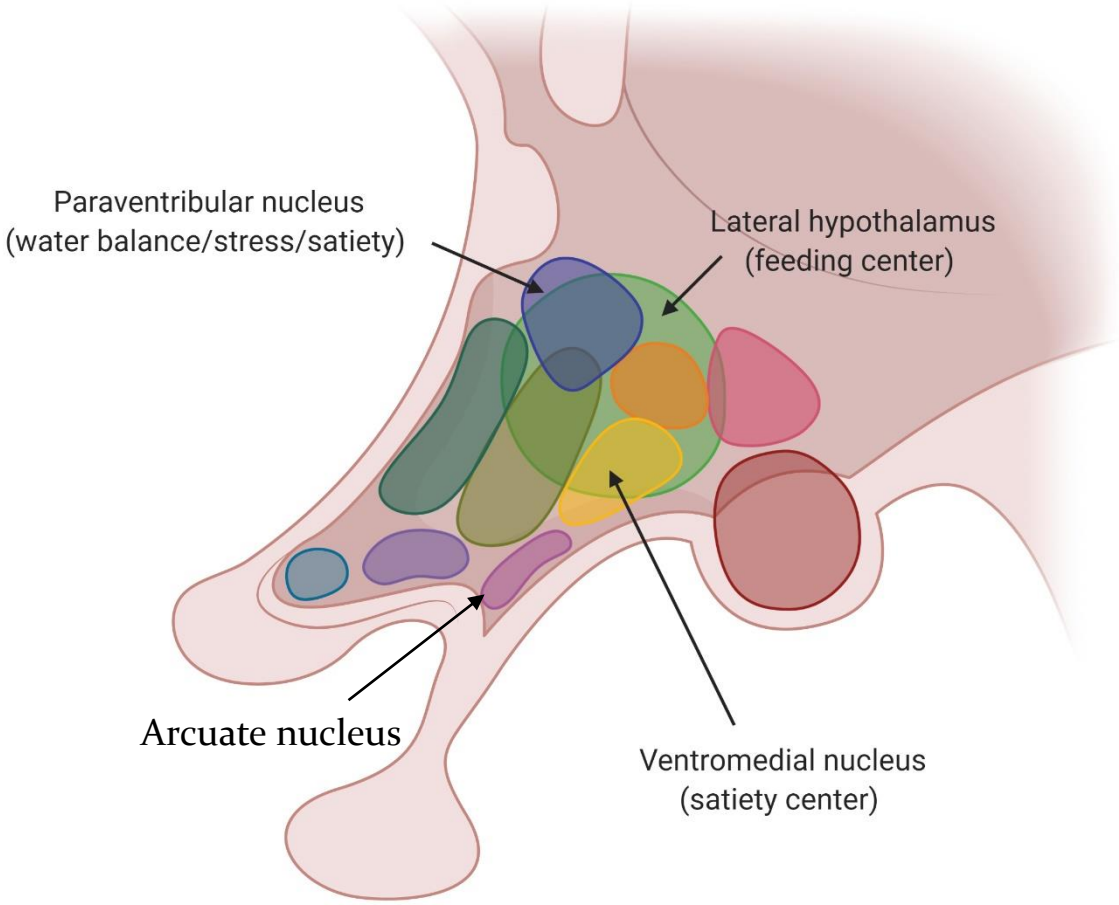
Control of food intake

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Neural centers of food intake



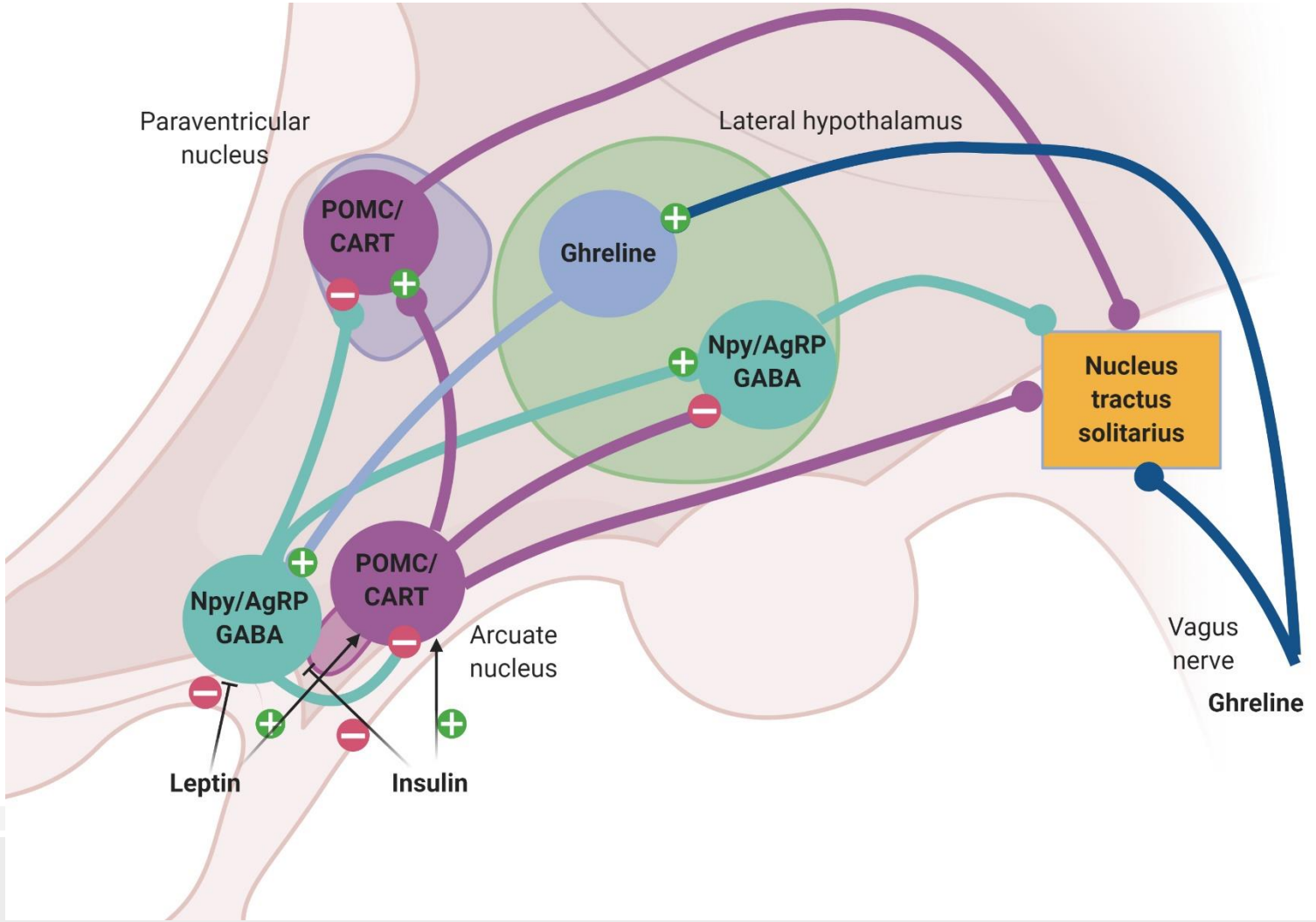
Signals influencing feeding/satiety centers

- Neural signals from GI tract (gastric distention)
- Chemical signals from nutrients in the blood (glucose, amino acids, fatty acids)
- GI hormones (CCK, ghrelin, PYY, insulin)
- Hormones from adipose tissues (i.e. leptin)
- Signals from cerebral cortex (sight, smell, taste)

Arcuate nucleus

- 2 types of neurons in arcuate nucleus that control feeding
- Pro-opiomelanocortin (POMC) neurons:□-melanocyte-stimulating hormone (α-MSH) and cocaine- and amphetamine-related transcript (CART) >> ↓ food intake and ↑ energy expenditure
- Neuropeptide Y (NPY) and agouti-related protein (AGRP) neurons >> ↑ food intake and ↓ energy expenditure

Arcuate nucleus



Regulation of food intake

- **Short-term regulation**: prevention of overeating at each meal
- **Long-term regulation:** maintenance of normal quantities of energy store in the body

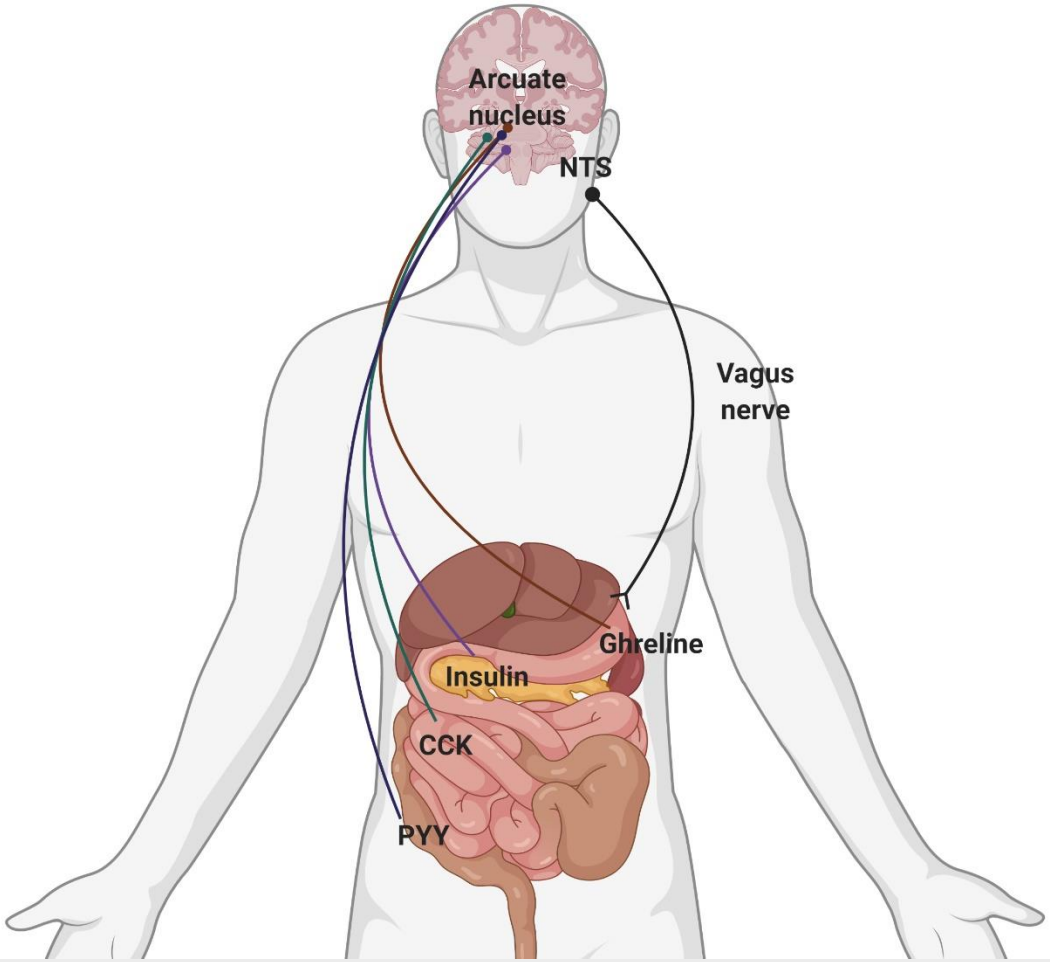
Short-term regulation

Short-term regulation-- signals

- GI filling: stretching of stomach and duodenum >> stretch inhibitory signals through vagus nerve >> inhibit feeding center
- GI hormones: CCK, insulin, GLP, peptide YY (PYY), ghrelin
- Oral receptors: chewing, swallowing, tasting inhibit feeding center (less intense and short life)

Hormonal regulation of food intake

- **Ghrelin, Peptide YY, CCK, insulin**



CCK

- Secretes in response to fat and protein duodenum
- Activates local sensory nerves in duodenum >> vagus nerve >> inhibit feeding center in hypothalamus
- Short-lived
- Only to prevent overeating during the meal

Peptide YY, insulin and GLP-1

- PYY: Secreted from the entire GI tract especially ileum and colon, high fat content is the strongest stimulator
- Insulin: secreted from pancreas in response to high glucose in the blood
- GLP-1: secreted in response to hyperglycemia and the presence of food in the intestines
- All 3 hormones inhibit food intake

Ghrelin – increase food intake

- **Structure**

28 amino-acid polypeptide, n-octanoyl contacts to 3rd serine (important for hormone activity)

Secreted by

- parietal cell in gastric fundus
- pancreas, intestine, hypothalamus and pituitary gland

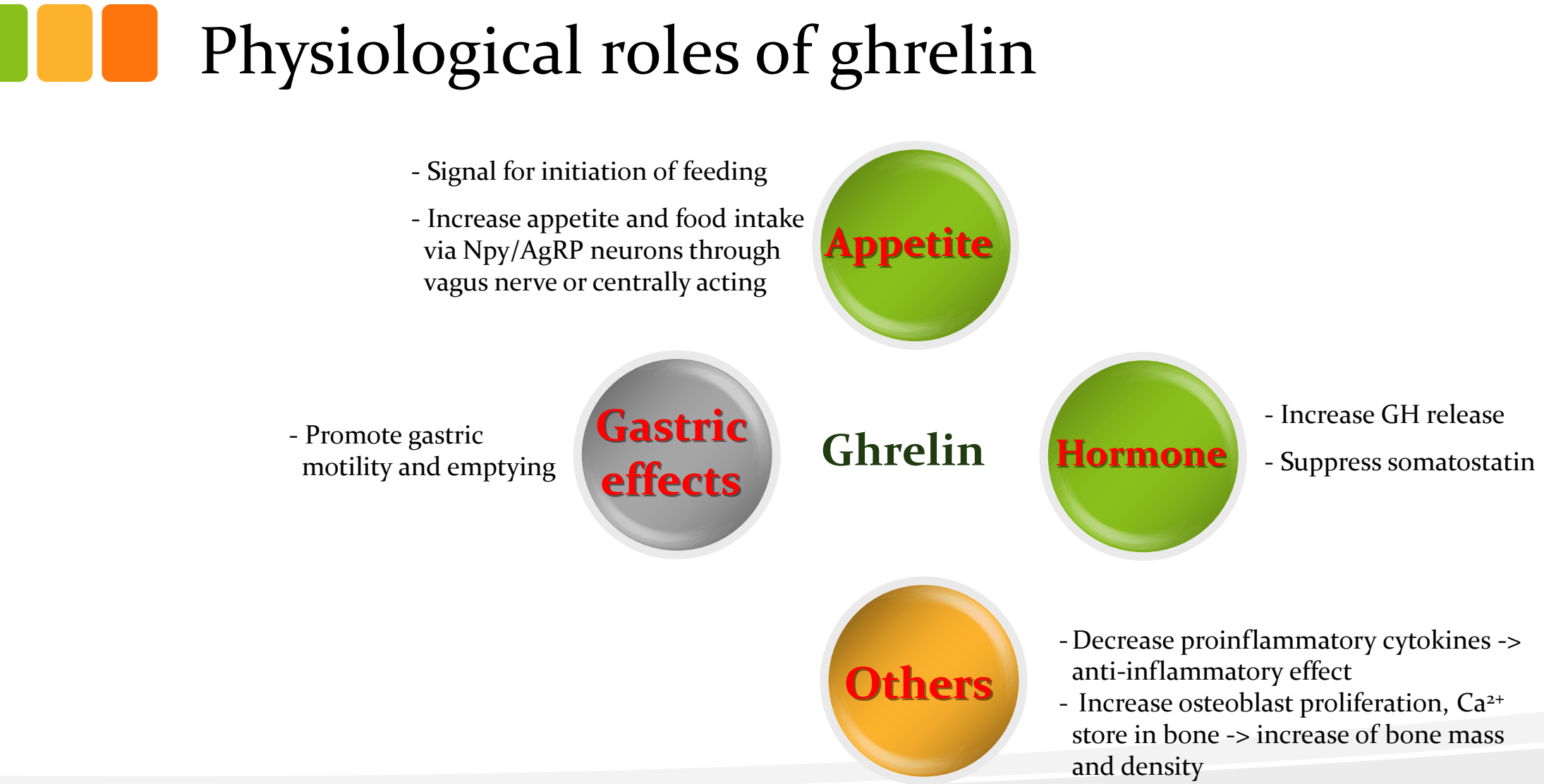
Bind to

Growth hormone-secretagogue-receptor 1a (GHS-R1a), a G-protein coupled receptor

Ghrelin levels

- Rise during fasting and peak right before meals and decrease after meals
- CHO can decrease ghrelin > fat

Physiological roles of ghrelin



Long-term regulation

Leptin

Secreted by

- Adipocytes (main)
- Small amounts from chief cells in stomach

Bind to

Janis kinase signal transduction and translation system (JAK STAT)

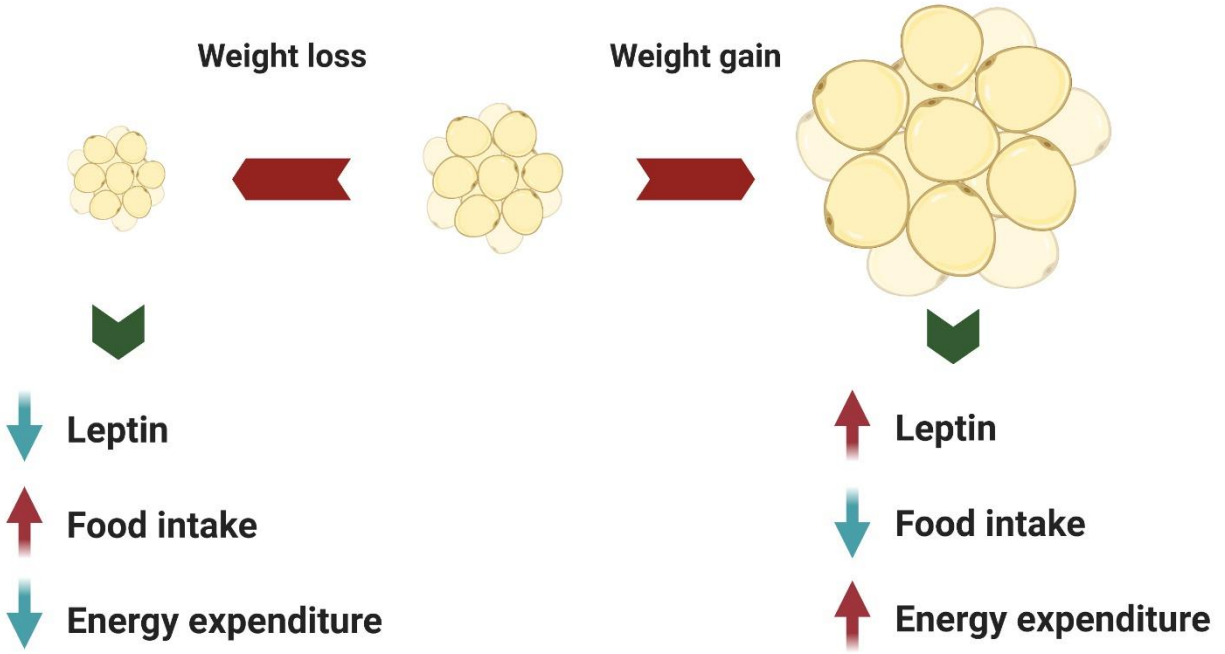
Leptin levels

- Reflect total body fat stores
- Increase in obesity (leptin resistance)

Leptin-- actions

- Inhibit NYP/AGRP neurons and activate POMC neurons >> reduce appetite and food intake
- Increase sympathetic nerve activities >> increase metabolic rate and energy expenditure
- Decrease insulin secretion >> lipolysis and decrease glycogen synthesis

Leptin and body weight changes





Thank you